The connected component of the local differential Galois group of the system

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned journal source file sources/sigma-2025-020-non-integrability-of-the-sasano-system-of-ty/pinned-original-source.tex (Non-Integrability of the Sasano System of Type D5(1) and Stokes Phenomena); no statement or proof step was rewritten, reordered or removed.

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Editorial note. The label gal-inf names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

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Editorial note. Journal numbering seed recomputed from the source itself; printed 1.22 is a cross-check only.

Editorial note. Revision r002: regenerated from the same spec after the r001 assembly crashed during packaging.

Editorial scope. the counted statement and its own proof environment only

Allowed standard prerequisites (editorial).

- Propositions for and ac of the same article. Supplies background that the delivered proof uses. The entry is declared as a dependency; it is not reproved on the card, and the load-bearing steps of the counted statement are carried out inside the delivered span.

Source reference. Original reference 2.5: sources/sigma-2025-020-non-integrability-of-the-sasano-system-of-ty/pinned-original-source.tex, source lines 377–377. Navigation pointer invoked at line(s) 1190: the counted text refers to this environment of the same journal unit, which is not retained in this card, so the reference

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Source reference. Original reference 3.5: sources/sigma-2025-020-non-integrability-of-the-sasano-system-of-ty/pinned-original-source.tex, source lines 956–956. Navigation pointer invoked at line(s) 1146: the counted text refers to this environment of the same journal unit, which is not retained in this card, so the reference is declared as a pointer carrying the number the article itself prints for it (the section and the place of the statement inside that section, derived from the shared section-scoped theorem counter of the source); the cited item is an allowed input and none of it is reproduced inside the span.

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Source reference. Original reference 3.1: sources/sigma-2025-020-non-integrability-of-the-sasano-system-of-ty/pinned-original-source.tex, source lines 651–651. Navigation pointer invoked at line(s) 1146: the counted text refers to this environment of the same journal unit, which is not retained in this card, so the reference is declared as a pointer carrying the number the article itself prints for it (the section and the place of the statement inside that section, derived from the shared section-scoped theorem counter of the source); the cited item is an allowed input and none of it is reproduced inside the span.

Theorem 1.22. *With respect to the formal and actual fundamental matrix solutions, given by Propositions* 3.1 *and* 3.5*, the connected component G^0 of the unit element of the local differential Galois group G at the origin of the system* (3) *is defined as follows:*

1. *If $\alpha \in \mathbb{N}$, then*

$$G^0 = \left\{ \begin{pmatrix} \lambda & 0 & 0 & \mu \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda^2 & 0 \\ 0 & 0 & 0 & \lambda^2 \end{pmatrix},\ \lambda \in \mathbb{C}^*,\, \mu \in \mathbb{C} \right\}.$$

2. *If $\alpha \in \mathbb{Z}_{\leq 0}$, then*

$$G^0 = \left\{ \begin{pmatrix} \lambda & \mu & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda^2 & 0 \\ 0 & 0 & 0 & \lambda^2 \end{pmatrix},\ \lambda \in \mathbb{C}^*,\, \mu \in \mathbb{C} \right\}.$$

3. *If* $\alpha \notin \mathbb{Z}$, *then*

$$G^0 = \left\{ \begin{pmatrix} \lambda & \mu & 0 & \nu \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda^2 & 0 \\ 0 & 0 & 0 & \lambda^2 \end{pmatrix},\ \lambda \in \mathbb{C}^*,\ \mu, \nu \in \mathbb{C} \right\}.$$

Proof. From Theorem 2.5, it follows that the local differential Galois group G at $\tau = 0$ of the system (3) is the Zariski closure of the group generated by the formal differential Galois group and the Stokes matrices St_π and St_0. Since when $\alpha \in \mathbb{Z}$ the formal monodromy $\hat{M}_0$ is equal to the identity matrix I_4 then the formal differential Galois group coincides with the exponential torus $\mathcal{T}$. Therefore, in this case G is generated by the exponential torus and the Stokes matrices St_π and St_0. Since when $\alpha \in \mathbb{N}$ the Stokes matrix St_π is equal to I_4 the Galois group G is generated by the exponential torus $\mathcal{T}$ and the Stokes matrix St_0. Hence for $\alpha \in \mathbb{N}$ the differential Galois group coincides with its connected component G^0 of the unit element and

$$G = G^0 = \left\{ \begin{pmatrix} \lambda & 0 & 0 & \mu \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda^2 & 0 \\ 0 & 0 & 0 & \lambda^2 \end{pmatrix},\ \lambda \in \mathbb{C}^*,\ \mu \in \mathbb{C} \right\}.$$

Similarly, when $\alpha \in \mathbb{Z}_{\leq 0}$ we have that G is generated topologically by $\mathcal{T}$ and St_π and

$$G = G^0 = \left\{ \begin{pmatrix} \lambda & \mu & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda^2 & 0 \\ 0 & 0 & 0 & \lambda^2 \end{pmatrix},\ \lambda \in \mathbb{C}^*,\ \mu \in \mathbb{C} \right\}.$$

When $\alpha \in \mathbb{Q}$ but $\alpha \notin \mathbb{Z}$, the formal differential Galois group is not connected since the group generated by $\hat{M}_0$ is not connected. However, in this case the connected component of the unit element of the formal differential Galois group coincides with $\mathcal{T}$. Therefore, in this case G does not coincides with G^0 but G^0 is generated by $\mathcal{T}$ and the Stokes matrices St_π and St_0. Hence

$$G^0 = \left\{ \begin{pmatrix} \lambda & \mu & 0 & \nu \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda^2 & 0 \\ 0 & 0 & 0 & \lambda^2 \end{pmatrix},\ \lambda \in \mathbb{C}^*,\ \mu, \nu \in \mathbb{C} \right\}.$$

In the last case when $\alpha \notin \mathbb{Q}$, we have that

$$G = G^0 = \left\{ \begin{pmatrix} \lambda & \mu & 0 & \nu \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda^2 & 0 \\ 0 & 0 & 0 & \lambda^2 \end{pmatrix},\ \lambda \in \mathbb{C}^*,\ \mu, \nu \in \mathbb{C} \right\}.$$

This ends the proof. □